

Siyuan Feng

BSc Mathematics, University of Warwick | Machine Learning Theory

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RESEARCH INTERESTS

Machine learning theory; optimization and training dynamics; hyperparameter transfer; scaling laws; learning-rate schedules.

EDUCATION

University of Warwick

Sep 2024 – Jun 2027 (expected)

BSc Mathematics

- Year 1 average: 82.5%; Year 2 average: 82.0% (First-Class level; GPA equivalent: 3.94/4.00).

RESEARCH EXPERIENCE

Researcher, Machine Learning Theory

Jul 2026 – Present

Coauthored with Prof. Fanghui Liu | Manuscript in preparation

From Normalization to Prediction: Warmup–Stable–Decay Schedule Transfer Across Training Steps

- Developed a theory of WSD schedule transfer across training-step counts in deep nonlinear MLPs. Introduced cumulative-learning-rate normalization to place different step counts in a common dynamical coordinate system, and derived a finite-step expansion of the complete WSD objective with a second-order remainder.
- Proved that one complete reference objective is insufficient in general, while two shorter-step reference objectives can predict the complete target WSD objective with second-order accuracy; also separated the normalization restriction gap and connected the prediction to wider μP models at fixed target step count.
- Extended the framework to optimizer scaling and dataset robustness: derived conditional power/near-power peak-learning-rate laws from objective geometry, identified explicit failure and curvature mechanisms, and proved deterministic bounds quantifying how budget-dependent dataset mismatch propagates into objective-prediction and target-performance error.

Warwick Undergraduate Research Support Scheme (URSS) Researcher

May 2026 – Jul 2026

Supervised by Prof. Fanghui Liu | Completed manuscript

Two-Step Optimal Learning-Rate Geometry under μP

- Derived an exact low-dimensional description of the two-step training dynamics and an explicit infinite-width formula for how final loss depends on the learning rate in a deep linear μP network.
- Proved that the optimal learning rate is unique for almost all datasets with a positive-definite Gram matrix, while also constructing full-rank datasets with exactly three equally good optimal learning rates.
- Showed that the optimal set remains stable under small changes to the data and that the corresponding local minima remain visible at finite width; completed a full research manuscript.

AWARDS & RESEARCH FUNDING

£500 Research Bursary — University of Warwick Undergraduate Research Support Scheme

2026

Awarded in support of the project *Two-Step Optimal Learning-Rate Geometry under μP* .

£75 Prize — 2025 Quant Championship

2025

Placed 4th out of 200 participants.

TECHNICAL SKILLS

- **Research computing** — Python (NumPy, SciPy, PyTorch, pandas, Matplotlib) and MATLAB: Built neural-network training pipelines, grid-based hyperparameter searches, simulations, finite-width checks, and reproducible figure-generation workflows.
- **Research workflow** — $\LaTeX$, Git, Linux, Jupyter, and Conda: Prepared mathematical manuscripts and maintained version-controlled, reproducible computational workflows.

INTERNSHIP, TEACHING & OUTREACH EXPERIENCE

University of Warwick – Undergraduate Mathematics Supervisor

Oct 2026 – Present

- Appointed and formally employed by the University of Warwick as an Undergraduate Mathematics Supervisor, contributing to the department's undergraduate teaching and academic supervision.

Tianma Microelectronics Co., Ltd. – Algorithm Engineering Intern

Aug 2025 – Sep 2025

- Built a pixel-level E-paper video ghosting simulation pipeline in MATLAB and Python, covering frame extraction, drive mapping, and grayscale output.
- Replaced pixel-wise loops with NumPy vectorization in a history-frame-difference algorithm, reducing single-run simulation time by approximately 85%.

A-Level Mathematics & Cambridge STEP Tutor

Apr 2024 – Aug 2024

- Delivered 100+ hours of mathematics tutoring; supported students who later received offers from Oxford, Imperial College London, NUS, and other universities.

Mathematics Content Creator, Bilibili

Ongoing

- Created 60+ original videos explaining calculus and advanced integration problems, reaching 12,000+ followers and 500,000+ cumulative views. Developed accessible problem-solving explanations for a broad online mathematics audience.